

### Advanced (Habanero)

**Q1.** Factor the expression  $x^2 + 5x + 4x + 20$  by grouping.

- (A)  $x(x + 5) + 4(x + 5)$
- (B)  $x(x + 4) + 5(x + 4)$
- (C)  $x(x + 5) + 4(x + 4)$
- (D)  $x(x + 4) + 5(x + 5)$
- (E)  $x(x + 5) + 4(x + 4)$

**Q2.** Factor the expression  $x^2 + 3x + 2x + 6$  by grouping.

- (A)  $x(x + 2) + 3(x + 2)$
- (B)  $x(x + 3) + 2(x + 3)$
- (C)  $x(x + 2) + 3(x + 3)$
- (D)  $x(x + 3) + 2(x + 2)$
- (E)  $x(x + 3) + 3(x + 2)$

**Q3.** Factor the expression  $x^2 - 4x - 5x + 20$  by grouping.

- (A)  $x(x - 4) - 5(x - 4)$
- (B)  $x(x - 5) - 4(x - 4)$
- (C)  $x(x - 4) - 5(x - 5)$
- (D)  $x(x - 5) - 4(x - 5)$
- (E)  $x(x - 4) - 4(x - 5)$

**Q4.** Factor the expression  $x^2 + 2x + 6x + 12$  by grouping.

- (A)  $x(x + 2) + 6(x + 2)$
- (B)  $x(x + 6) + 2(x + 6)$
- (C)  $x(x + 2) + 6(x + 6)$
- (D)  $x(x + 6) + 2(x + 2)$
- (E)  $x(x + 6) + 6(x + 2)$

**Q5.** Factor the expression  $x^2 - 3x - 2x + 6$  by grouping.

- (A)  $x(x - 2) - 3(x - 2)$
- (B)  $x(x - 3) - 2(x - 3)$
- (C)  $x(x - 2) - 3(x - 3)$
- (D)  $x(x - 3) - 2(x - 2)$
- (E)  $x(x - 3) - 3(x - 2)$

**Q6.** Factor the expression  $x^2 + x + 4x + 4$  by grouping.

- (A)  $x(x + 1) + 4(x + 1)$
- (B)  $x(x + 4) + 1(x + 4)$
- (C)  $x(x + 1) + 4(x + 4)$
- (D)  $x(x + 4) + 1(x + 1)$
- (E)  $x(x + 4) + 4(x + 1)$

**Q7.** Factor the expression  $x^2 - 2x - 5x + 10$  by grouping.

- (A)  $x(x - 5) - 2(x - 5)$
- (B)  $x(x - 2) - 5(x - 2)$
- (C)  $x(x - 5) - 2(x - 2)$
- (D)  $x(x - 2) - 5(x - 5)$
- (E)  $x(x - 2) - 2(x - 5)$

**Q8.** Factor the expression  $x^2 + 6x + 2x + 12$  by grouping.

- (A)  $x(x + 2) + 6(x + 2)$
- (B)  $x(x + 6) + 2(x + 6)$
- (C)  $x(x + 2) + 6(x + 6)$
- (D)  $x(x + 6) + 2(x + 2)$
- (E)  $x(x + 6) + 6(x + 2)$

#### Open Ended Questions (FRQ)

**Q9.** Factor the expression  $x^2 + 4x + 3x + 12$  by grouping and verify by multiplication.

**Q10.** Factor the expression  $x^2 - 5x - 4x + 20$  by grouping and verify by multiplication.

#### Chef's Tips

- Check for the common binomial factor after grouping.
- Use the distributive property to verify factoring.
- Check for any remaining factoring opportunities.